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High-unblocking-temperature haematite magnetizations of Late Palaeozoic red beds from the Okcheon zone, southern part of the Korean Peninsula

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SUMMARY

The Late Carboniferous and Permo-Triassic red beds of the Pyeongan Supergroup in the Okcheon zone, southern part of the Korean Peninsula, were found to contain three components of remanence. These components, which are resolved by thermal demagnetization, are characterized by high (Hhc for Late Carboniferous and Hhpc for Permo-Triassic), intermediate (Hm) and low (Hl) unblocking temperatures. As the maximum unblocking temperature spectra of the Hhc (Hhpt) component often overlapped the Hm spectra above 680 °C, 3 ° to 5 °C steps were needed above 670 °C to resolve the Hhc-(Hhpt-) component magnetization.

The tilt-corrected mean direction of the Hhc component red beds is $D=255.5°$, $I=-2.4°$ ($\alpha_{95}=20.8°$), and that of the Hhpt component is $D=271.9°$, $I=-33.4°$ ($\alpha_{95}=16.5°$). The precisions of mean directions of Hhc and Hhpt components improve after tilt correction and allow the directions to be distinguished from each another. This indicates that both Hhc and Hhpt component magnetizations were acquired during or shortly after the deposition of the red beds.

According to micropetrographic observations, the haematite in the red beds can be divided into three types: detrital specularite (A), non-detrital specularite (B), and pigmental haematite (C). Type A grains are tens of micrometres in size, and many of them are martite. The remanent magnetization of type A is possibly depositional remanent magnetization (DRM), post-DRM or chemical remanent magnetization (CRM) acquired close to the deposition time, whereas that of types B and C is of CRM origin acquired during and after folding, respectively. The time of folding is well established as a Triassic to Jurassic event. We believe that the remanent magnetization carried by type A grains is primary, and is correlated to Hhc-(or Hhpt-) component magnetization.

Key words: haematite, magnetization.

1 INTRODUCTION

Haematite is a common magnetic mineral in rocks of all ages, and its remanent magnetization is stable at high temperature and under oxidizing conditions. Haematite has contributed to the great success of palaeo-reconstruction in many palaeomagnetic studies (e.g. Creer 1962; Douglass 1988; Channell, McCabe & Woodcock 1992). It has been debated whether the remanent magnetization in red beds is depositional or chemical. Late Palaeozoic remagnetization of CRM origin is well known in the Appalachian red beds (Kent & Opdyke 1985; Stearns & Van der Voo 1987). However, palaeomagnetic results from many other red beds suggest that the remanent magnetization of haematite was acquired during or close to the time of deposition as DRM or post-DRM (e.g. Douglass 1988).

Some of the primary directions of haematite have been obtained by multicomponent analysis. However, errors can be introduced in such an analysis if thermal demagnetization is not sufficiently detailed to separate components with the overlapping unblocking temperature spectra. Roy & Park (1972) claimed that thermal demagnetization could not separate a DRM from a CRM due to specularite (CRM_A), even though it could remove a CRM due to red pigment (CRM_B).

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From petrographical observations, Walker, Larson & Hoblitt (1981) suggested that most haematite grains in the Moenkopi Formation, Colorado Plateau were formed by authigenesis resulting principally from intrastratal breakdown of iron-bearing detrital grains, and thus the haematite remanent magnetizations were acquired as CRM. Only a small proportion of haematite was of detrital origin, giving DRM or post-DRM. Dekkers & Linssen (1989) investigated rock-magnetic properties of the multiphase mineralization of haematite. They found that haematite produced chemically at low temperature lacked sharp Morin transitions. However after annealing at 700 °C, such haematite showed a sharp Morin transition due to the removal of the defect moment.

Palaeozoic rocks in the Okcheon zone have been reported to be remagnetized after the Palaeozoic (e.g. Shibuya *et al.* 1988). Some of the *in situ* secondary components with high unblocking temperatures had directions identical with the geocentric axial dipole field direction. Several palaeomagnetic poles from the Late Carboniferous and Permo-Triassic red beds in the Okcheon zone were obtained by Shibuya *et al.* (1988), Kim & Van der Voo (1990), Kim, Lee & Choi (1992), and Doh & Piper (1994). Shibuya *et al.* (1988) suggested that the Okcheon zone originally belonged to the South China Block, based on palaeopoles. Kim & Van der Voo (1990) and Kim *et al.* (1992) argued that their palaeopoles were different from those of the South and North China Blocks obtained by Lin, Fuller & Zhang (1985). Recently, Doh & Piper (1994) reported a wide spread in the declinations of remanent directions from the syncline of the Okcheon zone, and interpreted these results as showing that the synclinal zone is made up of complicated rotating blocks. These arguments about the pre-Jurassic palaeopole positions from the Okcheon zone, which are crucial in understanding the tectonics of East Asia, are not resolved.

We believe that the characteristic directions cannot be completely reliable without careful rock magnetic analyses. We studied Late Carboniferous and Permo-Triassic red beds in the Okcheon zone in detail. The multicomponent magnetizations were resolved by detailed thermal demagnetization, and verified by careful palaeomagnetic, rock magnetic, and petrographic analyses. This paper focuses on the verification of the existence of the multicomponent magnetizations containing characteristic remanences from Late Palaeozoic red beds of the zone. Tectonic implications for the evolution of East Asia based on our new palaeomagnetic results will be presented in a subsequent paper.

2 GEOLOGICAL SETTING AND PALAEOMAGNETIC SAMPLING

The NE–SW trending Okcheon zone is about 80 km wide and is located in the middle part of the Korean Peninsula (Fig. 1). It separates the Yeongnam and the Gyeonggi Precambrian blocks, and is one of the prominent structural features of East Asia. The zone is divided into a northeastern non-metamorphic and a southwestern metamorphic part.

The northeastern Okcheon zone, the sampling area, contains the only Palaeozoic sedimentary outcrops, made up of the Cambrian to Early Silurian Joseon Supergroup and the Late Carboniferous to Permo-Triassic Pyeongan Supergroup. Both supergroups, which are separated by a paraconformity, were folded during the Songrim (Triassic, equivalent to the Indosinian) and the Daebo (Jurassic) Orogenies (Lee 1987). The groups show lithological variations and several type areas have been established. The biostratigraphies of the Duwibong and the Yeongweol types have been well established in the Baegunsan and Yeongweol areas respectively.

2.1 The Baegunsan area

In the southern part of the Duwibong type area, the Baegunsan Syncline is more than 100 km long and about 15 km wide (Fig. 1a). The Manhang Formation, the lowermost part of the Pyeongan Supergroup, is characterized by sandstones and shales of red or greenish-grey colour, intercalated with pebbly conglomerate and limestone layers. Its stratigraphic age has been suggested, on the basis of *fusulina* fossils, to be Early Moscovian (Lee 1987), which can be dated at 311 to 307 Ma, according to Harland *et al.* (1989).

The Donggo Formation, the uppermost part of the Pyeongan Supergroup, consists mainly of pale green sandstone and some intercalated red beds. A latest Permian or earliest Triassic age has been suggested, because plant fossils of latest Permian age were found in the Gohan Formation below the Donggo Formation (Lee 1987).

Three red-bed sites (Sites 615, 617, and 927) and one pale-grey sandstone site (Site 633) were sampled from the lower part of the Manhang Formation. Five sites (Sites 634, 923, 924, 925, and 926) of the interbedded red beds and green sandstones were sampled from the Donggo Formation (Fig. 1a and Table 1).

2.2 The Yeongweol area

This area is located about 40 km west of the Baegunsan sampling sites (Fig. 1b). A large thrust is located to the west of the sampling sites. The formations extend longitudinally by about 20 km. The Yobong Formation, the lowermost part of the Pyeongan Supergroup, is characterized by red to greenish-grey sandstones and shales intercalated with pebbly conglomerate and limestone layers (Lee 1987). Deposition of the Yobong Formation occurred during the late Bashkirian to Moscovian. Although the stratigraphy and the geological structure of the Yobong Formation are different from those of the Manhang Formation in the Baegunsan area, the two formations are generally thought to be coeval. Four red-bed sites (Sites 604, 605, 607 and 608) and one limestone site (Site 606) were sampled from the Yobong Formation (Fig. 1b and Table 1).

3 LABORATORY PROCEDURES

Seven to nine hand-size block samples were collected at each sampling site, covering 1–4 m stratigraphic thickness, which may decrease the chance of recording a transient feature of the past geomagnetism. All samples were oriented by using a magnetic compass. Specimens 24 mm in diameter and 22 mm long were cored from each block.

Natural remanent magnetization (NRM) of the specimens was measured using a two-axis cryogenic rock magnetometer (ScT C-112). Most specimens acquired a significant viscous

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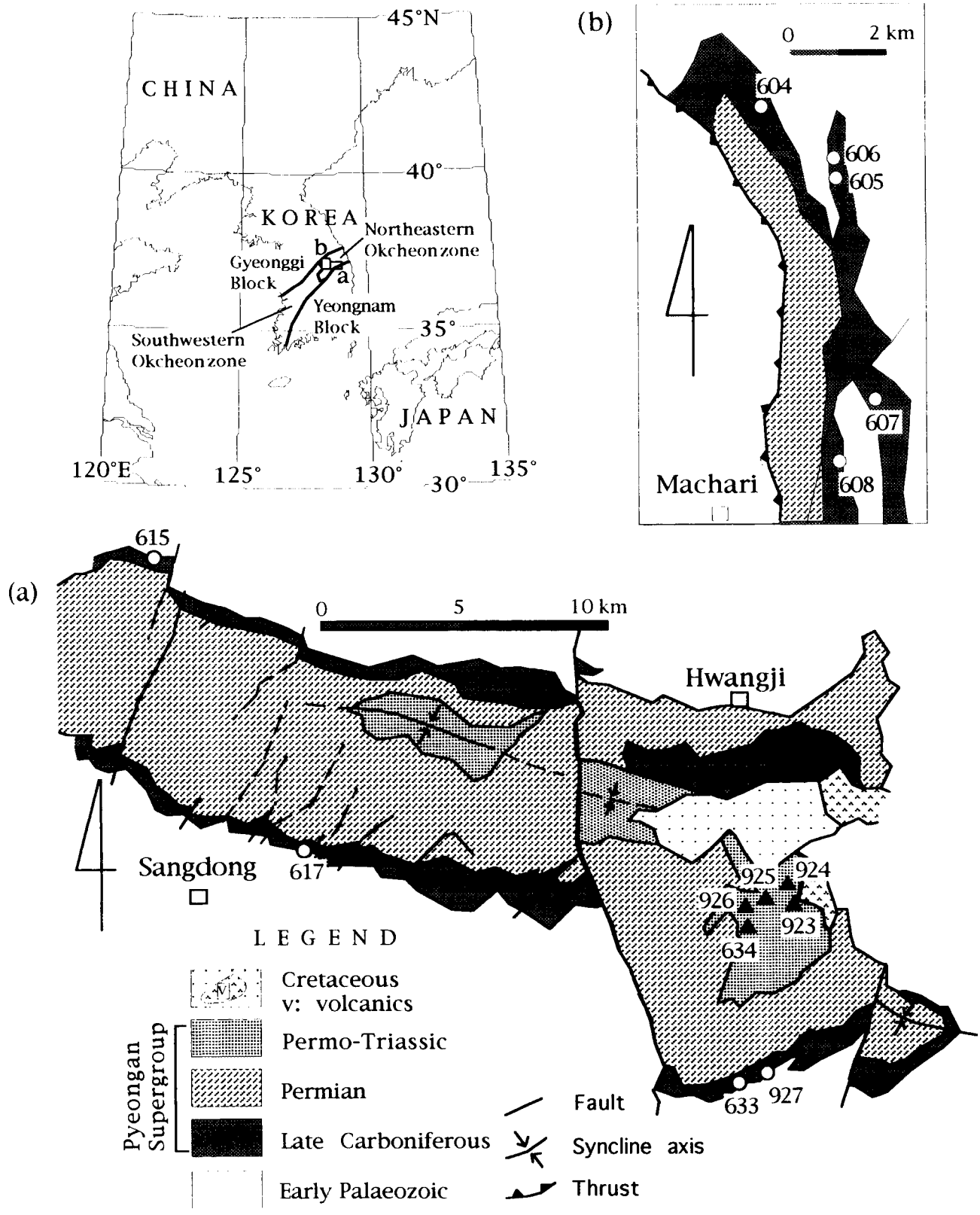


Figure 1. Sample localities in the Baegunsan synclinal area (a) and the Yeongweol area (b). Open circles and solid triangles denote sample site of Late Carboniferous and Permo-Triassic sedimentary rocks, respectively. The base map is modified from Lee (1987).

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Table 1. Stratigraphy of the Pyeongan Supergroup. (Numbers next to formation names indicate sampling sites.)

Period	Epoch Harland et al. (1989)	Baegunsan area Lee (1987)	Yeongweol area Lee (1987)
Triassic			
		Donggo 634,923,924 925,and926	
	245 (Ma)		
Permian	Zechstein	(Paraconformity)	
		Gohan	
		Dosagok	
	256	Hambaeksan	
		Jangseong	Mitan
	Rotliegendes	Bamchi	Bamchi
	290		
Carboniferous Pennsylvanian	Gzelian	(Paraconformity)	(Paraconformity)
	295		
	Kasimovian		
	303	Gumcheon	Pangyo
	Moscovian	Manhang 615,617, 633,and927	Yobong 604,605,606, 607,and608
	311		
	Bashkirian		

remanence when inserted through ambient fields into the measuring space of the magnetometer, where the residual field was less than 20 nT. We developed a sample handling tool to rotate specimens about a horizontal axis at the measuring position in the low-field space. Using this handler, exposure of specimens to the laboratory field is avoided during measurements. Viscous decay was studied with a pen-recorder connected to the magnetometer. A time of at least 20 min, sometimes 1 hr, was needed to minimize viscous remanence in specimens heated above 660 °C.

All specimens were demagnetized thermally in a non-inductive electric furnace enclosed in four nested shields. The residual field in the furnace during the cooling cycle was about 5 nT. Although the number of demagnetization steps differed from sample to sample, 3° to 5 °C steps were employed to delineate unblocking temperature components above 670 °C. Low-field susceptibility was measured using a Bartington MS2 susceptibility meter after each step of the progressive thermal demagnetization in order to detect chemical changes during the heating. Alternating demagnetization was not used because the maximum field of our alternating demagnetization equipment is 100 mT, obviously too low to reveal multiple components of NRM of the red-bed samples.

The progressive-thermal-demagnetization data were analysed using the principal component analysis method (Kirschvink 1980) and also through the combined analysis of great circles and directly observed remanent magnetizations (McFadden & McElhinny 1988). Spectral diagrams of the unblocking temperature were also helpful in separating the components. Site and formation means were calculated using Fisher's statistics (Fisher 1953).

Thermomagnetic analysis, isothermal remanent magnetization (IRM) acquisition, and thermal demagnetization of composite IRMs (Lowrie 1990) were applied and studied in order to identify magnetic minerals that carry the remanent magnetization. For composite IRMs, DC fields of 1.3 T, 0.4 T, and 0.12 T were imparted as hard, medium, and soft IRM components. Thermomagnetic analysis was performed using an automatic recording semi-horizontal thermobalance. Optical micropetrographic inspection of the polished surfaces of the specimens and thin sections was carried out under reflected and transmitted light.

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4 PALAEOMAGNETIC ANALYSIS

4.1 High-unblocking-temperature magnetizations observed at Sites 617 and 615 in the Late Carboniferous Manhang Formation

4.1.1 Site 617

Site 617 is located on the southern limb of the Baegunsan synclinal zone, while Site 615 is on the northern limb (Fig. 1a). These two sites make a pair for the fold test (Graham 1949), as will be shown later. Seven block samples (A to G in stratigraphically ascending order) were taken at Site 617, but the high fissility of shaley samples (C and E) made it impossible to core specimens in the laboratory. Thus, only specimens of the fine-grained red sandstone from five stratigraphic positions of the outcrop were measured. NRM intensities of these specimens range from 1.8 to 6.5×10^{-2} A m^{-1}, higher than average for sedimentary rocks.

The typical thermomagnetic curve (Fig. 2a) suggests a pre-

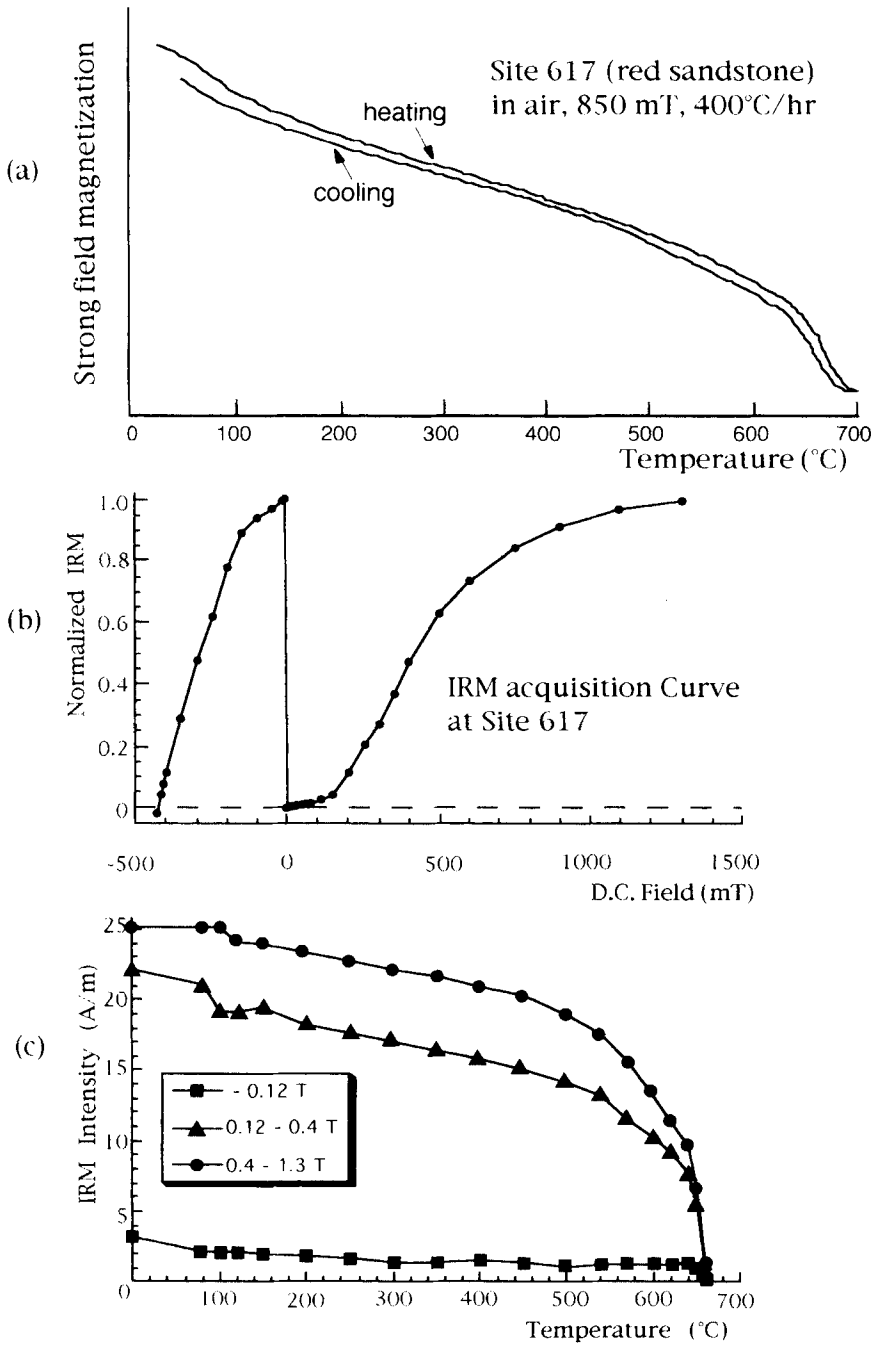


Figure 2. (a) Thermomagnetic curve. (b) IRM and back-field aquisition curves. Remanence acquisition coercivity is 425.2 mT. (c) Thermal demagnetization of composite IRMs (Lowrie 1990). Slight drops in hard and medium IRMs at about 100 °C indicate the presence of goethite. All diagrams indicate that haematite is the main magnetic carrier at Site 617.

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dominance of haematite as the magnetic mineral in the samples. The IRM failed to saturate in the applied DC fields of 1.3 T (Fig. 2b), indicating the dominance of high-coercivity minerals, such as haematite and goethite. The presence of haematite and goethite is clearly shown by the thermal demagnetization curves of composite IRMs (Fig. 2c). Magnetite, maghemite, pyrrhotite, and greigite were not detected. The absence of those thermally less-stable minerals is also supported by the lack of an abrupt change in low-field susceptibility upon heating below 600 °C (Fig. 3). Considerable changes in the low-field susceptibility are observed upon heating above 600 °C. These are possibly caused by the production of ultrafine haematite grains (superparamagnetic) at high temperatures (e.g. Dekkers & Linssen 1989). If minerals such as maghemite or pyrrhotite exist, low-field susceptibility would be expected to increase above 400 °C (e.g. Torii *et al.* 1992). Production of superparamagnetic grains upon heating can also be inferred from the prolonged viscous decay (several tens of minutes) observed during remanent magnetization measurements.

Throughout the progressive thermal demagnetization, we found two or three component magnetizations in all specimens. The demagnetization results are shown in Fig. 4, which contains vector demagnetization diagrams, normalized thermal decay curves for NRM, and unblocking spectra in the temperature range above 500 °C. Three specimens (D, F, and G) contain a large low-unblocking-temperature component, while two other specimens (A and B) contain no low-unblocking-temperature component, as shown by block-shaped thermal decay curves (Fig. 4b). The first three specimens lost half of their original NRM intensities upon heating at 600 °C. Below 600 °C, the linear segment of the vector end-points is well defined, as shown by the typical specimen G. This component of relatively low unblocking temperature may be carried by the haematite. Goethite makes little contribution to remanent magnetization, as suggested by the absence of change in intensity below 200 °C (See 617A in Fig. 4b). The direction carried by the component below 600 °C was isolated following Kirschvink's (1980) method, and is identical to the present geocentric axial dipole field direction (I = 56.5°) as shown in Fig. 5. This component, with a maximum unblocking temperature of about 600 °C, is aligned parallel to the geocentric axial dipole field direction, and is hereafter referred to as the Hl component.

Vector endpoints on the demagnetization diagrams have curvilinear trends in the high temperature range (Fig. 4a), suggesting the presence of at least two magnetization components in that range. The directions from curvilinear segments coincide with a great-circle trajectory, as illustrated in Fig. 5, indicating the simultaneous removal of two components of remanent magnetization. As shown in this figure, the great circles meet at a point which represents the characteristic direction of the site. The linear fitting method (Kirschvink 1980) is applied to three or four vector endpoints in the highest temperature range, giving the high-temperature (HT) directions in Fig. 5 (squares A, B, D, F, and G). These HT directions do not coincide with the characteristic direction defined by the intersection of great circles. This implies incompleteness of the thermal demagnetization and/or considerable overlapping of unblocking temperatures of the lower- and higher-temperature components. We hereafter call the characteristic direction with the highest unblocking temperature the Hhc component. The maximum unblocking temperature is estimated, on the basis of unblocking temperature spectra, to exceed 680 °C (Fig. 4c).

The Hm component with intermediate unblocking temperature is defined as the magnetization component that overlaps the Hhc. The maximum unblocking temperature of the Hm component sometimes coincides with that of the Hhc component. The linear fitting method (Kirschvink 1980) is applied to more than three vector endpoints in the intermediate temperature range, giving the directions shown in Fig. 5. The intermediate-temperature (MT) magnetizations (inverse

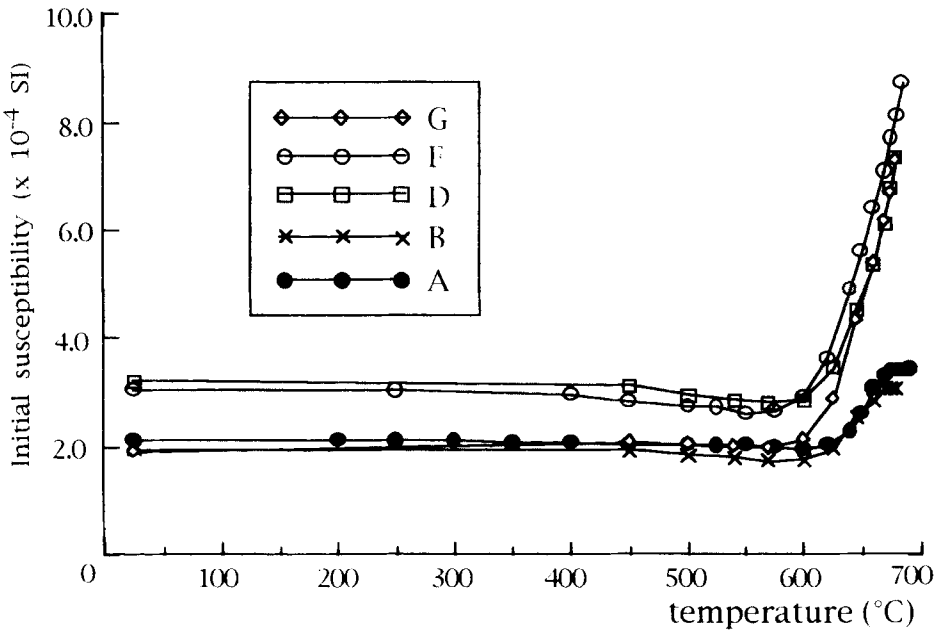


Figure 3. Low-field susceptibility measured after each step of the thermal demagnetization. A to G correspond to block samples of Site 617 (see text for relative position of block samples).

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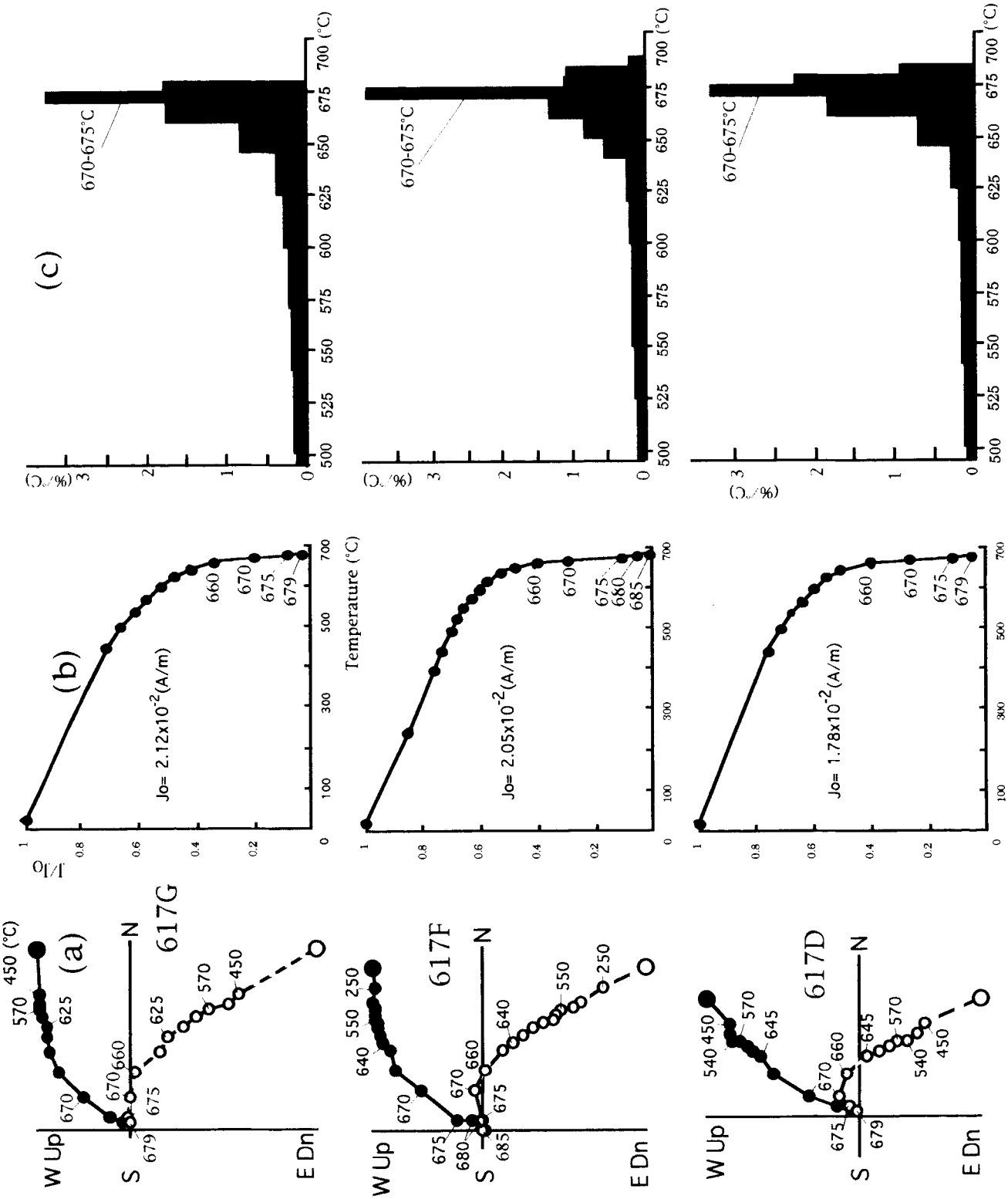


Figure 4. Thermal demagnetization behaviour for specimens at Site 617. (a) Demagnetization vector diagrams (*in situ*). Solid and open circles indicate vector endpoints projected on the horizontal and vertical planes, respectively. (b) Normalized thermal demagnetization diagrams. (c) Unblocking temperature above 500 °C. Note that the spectra peak at 670 °–675 °C.

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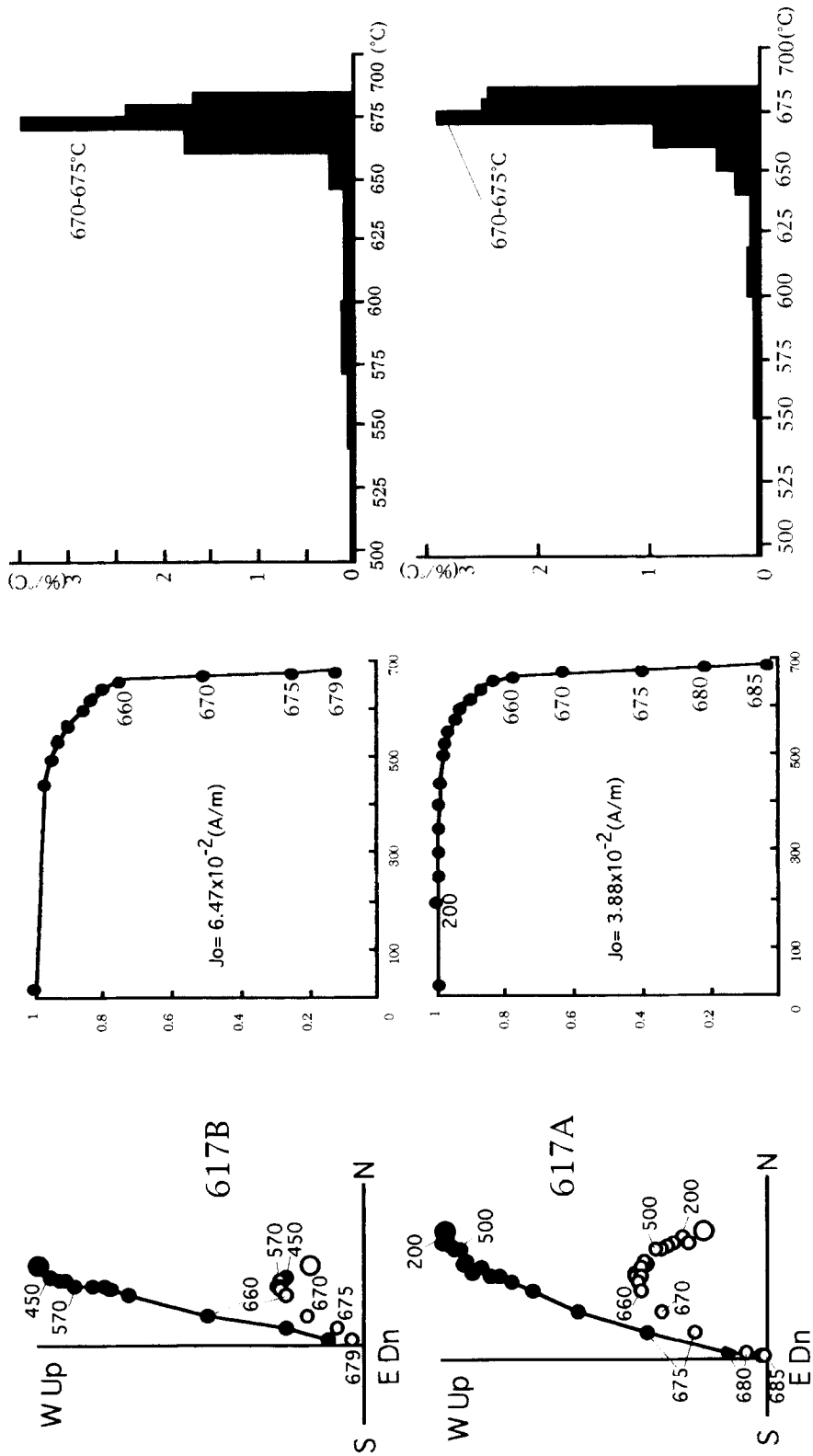


Figure 4. *(Continued.)*

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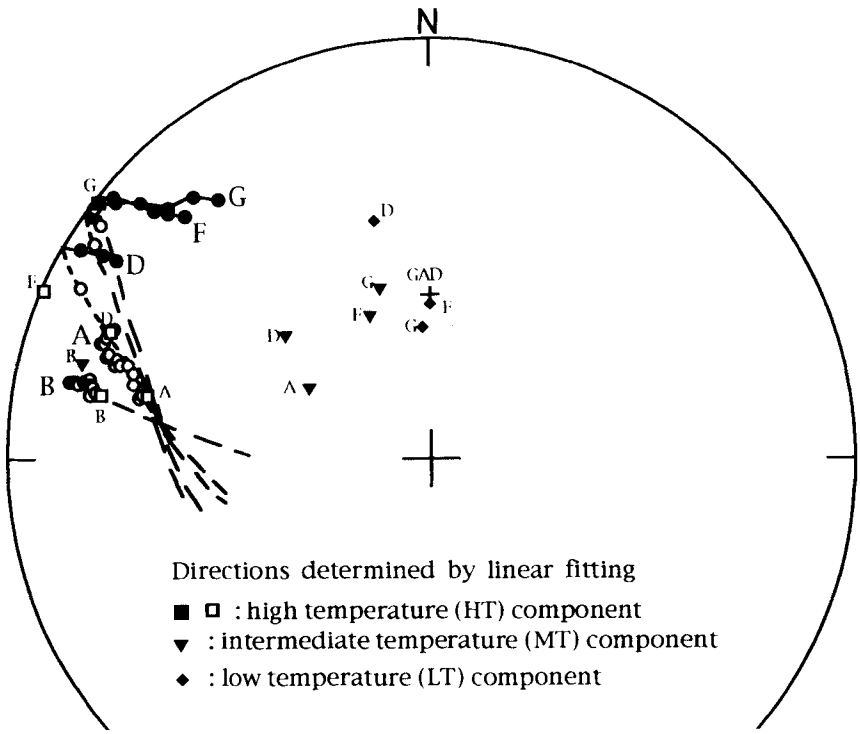


Figure 5. Great-circle trajectories of specimens from Site 617. Solid and open symbols are projections on the lower and upper hemisphere, respectively. A cross (GAD) indicates the geocentric axial dipole direction in the sampled area. The demagnetization temperature ranges from the great circle (Gc) and for each curvilinear segment (HT, MT, and LT) are: A: 450°–680°C (Gc), 675°–685°C (HT), 200°–500°C (MT). B: 450°–679°C (Gc), 670°–679°C (HT), 450°–570°C (MT). D: 600°–675°C (Gc), 670°–679°C (HT), 570°–645°C (MT), 450°–540°C (LT). F: 600°–670°C (Gc), 670°–685°C (HT), 550°–640°C (MT), 250°–550°C (LT). G: 600°–675°C (Gc), 670°–679°C (HT), 570°–625°C (MT), 450°–570°C (LT).

triangles A, B, D, F, and G in Fig. 5) are scattered. The F and G directions are similar to the geocentric axial dipole field direction (cross GAD in Fig. 5), but those of A, B, and D differ widely from one another. We interpret the MT direction at Site 617 as representing the Hm component and believe that it was acquired during a syn- or post-tectonic process. The low-temperature (LT) directions (diamonds D, F, and G in Fig. 5), interpreted as being the Hl component, are similar to the geocentric axial dipole field direction (cross GAD in Fig. 5). These magnetizations may have been acquired during the Brunhes Epoch.

4.1.2 Site 615

Site 615 is located on the northern limb of the Baegunsan synclinal zone (Fig. 1a). Eight block samples (spanning 3 m thickness) were taken from the Moscovian Manhang Formation.

Typical examples of thermal demagnetization results are given in Fig. 6(a)–(c). A high-unblocking-temperature component is observed as a linear segment above 660°C on the vector demagnetization diagram in Fig. 6(a) (specimen F). We regard this direction, with a southerly declination and shallow upward inclination (square F in Fig. 6d), as the characteristic direction. The remaining three specimens, C, G, and H, yield identical directions (Fig. 6d). In Fig. 6(b) (specimen E), the vector endpoints on the demagnetization diagram between 600° and 679°C display a curvilinear trend and lie along a great-circle trajectory (trajectory E in Fig. 6d) when plotted on the equal-area projection. This indicates a simultaneous removal of two components. The lower-temperature component is determined from the 450° to 600°C linear segment in Fig. 6(b) (triangle E in Fig. 6d), and is characterized by a vertical inclination (*in situ*). Although the higher-temperature component was not directly observed, the great-circle trajectory (trajectory E in Fig. 6d) moves toward a stable component as observed in specimen G (square G in Fig. 6d). As shown in Fig. 6(c) (specimen B), a component is isolated between 450° and 570°C, and another between 625° and 679°C. The direction of the lower-temperature component is identical to the geocentric axial dipole field direction (diamond B, Fig. 6d). The higher-temperature component in specimen B has a near-vertical inclination (triangle B in Fig. 6d). The same magnetization direction is observed for the lower component in specimens E (450°–600°C in Fig. 6b) and F (450°–540°C in Fig. 6a). Therefore, on the basis of maximum unblocking temperature, the magnetization is divided into three components as follows: a high-unblocking-temperature (Hhc) component with a shallow, upward inclination (squares C, F, G, and H in Fig. 6d); an intermediate-unblocking-temperature (Hm) component with a steep inclination (triangles A, B, E, and F in Fig. 6d); and a low-unblocking-temperature (Hl) component with a direction similar to that of the geocentric axial dipole field (diamonds A, B, and C in Fig. 6d).

In normalized thermal decay curves for NRM, the removal of the Hhc component is shown by a considerable drop of

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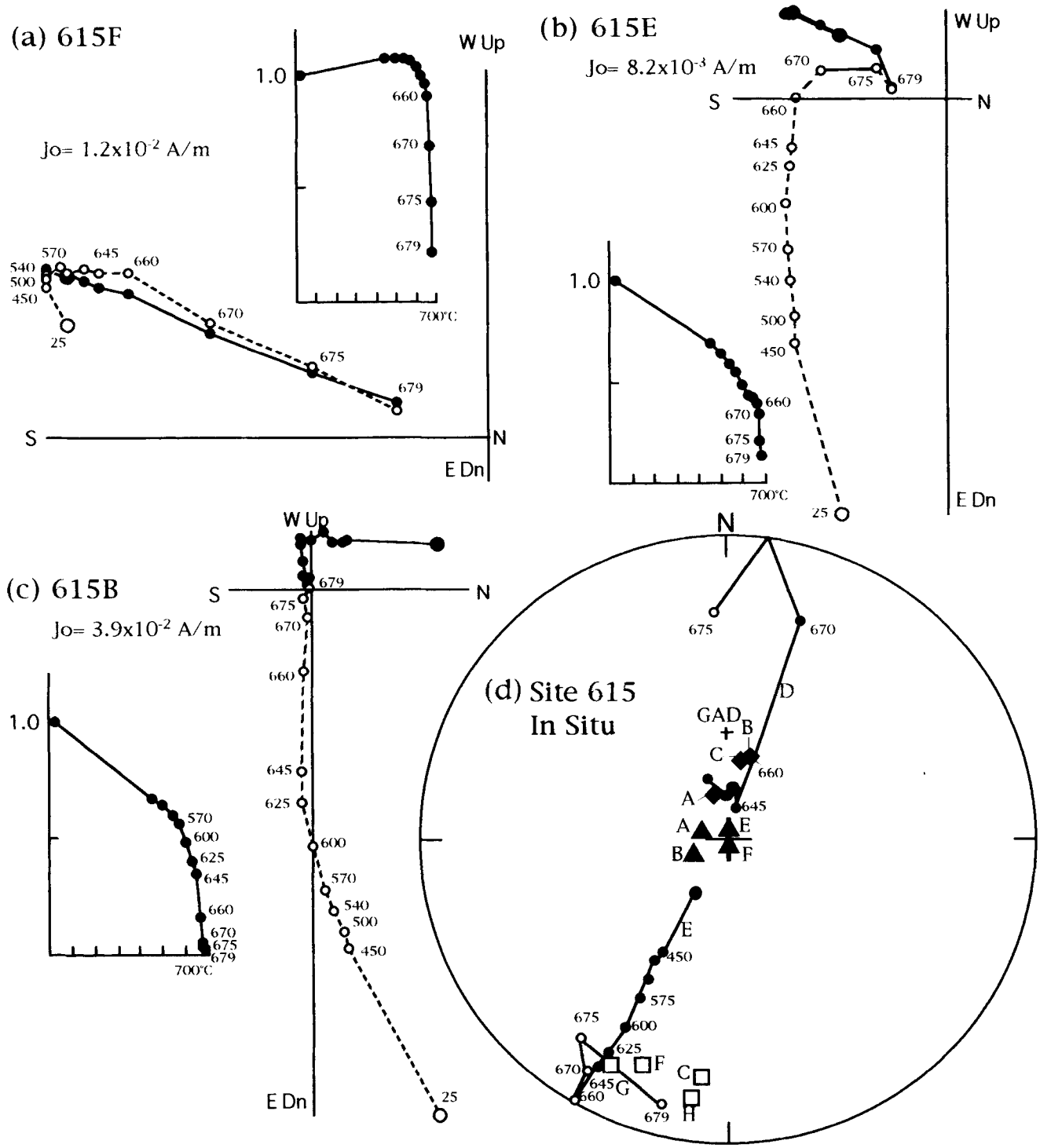


Figure 6. Thermal demagnetization result for Site 615F (a), E (b), and B (c). For explanation of symbols see Fig. 4 (a). (d) Equal-area projection. Diamonds, triangles, squares, and crosses indicate Hs, Hm, Hh, and the geocentric axial dipole field directions, respectively. Solid and open symbols are projections on the lower and upper hemisphere, respectively. All directions are shown before tilt correction.

intensity in the range 660 °–679 °C (Figs 6a, b). In Fig. 6(c), the intensity drop indicates Hm removal in the range 600 °–679 °C. This indicates that the distribution of unblocking temperatures of the Hhc component is very narrow, while that of the Hm component is relatively broad. The mean direction of the Hhc component at Site 615 is calculated using combined analysis (McFadden & McElhinny 1988) of the two great circles (great-circle trajectories D and E in Fig. 6d) and four direct observations (squares C, F, G, and H in Fig. 6d). This direction is listed in Table 2.

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Table 2. Summary of palaeomagnetic results from Late Carboniferous (upper) Permo-Triassic (lower) red beds in the northeastern Okcheon zone.

Formation	Site	Long(°E)/Lat(°N)	nsp(CPA+Gc)	D'	I'	α'95	k1	D	I	α95	k2
Manhang	615	128.78/37.23	6(4+2)	199.2	-21.3	10.3	46.9	234.9	-21.3	11.0	40.3
	617	128.88/37.16	5(0+5)	281.2	-37.0	6.5	293.4	269.8	-15.6	6.6	291.2
	927	129.07/37.10	4(4+0)	242.7	20.3	8.4	119.4	257.6	17.1	8.4	119.5
Yobong	605	128.48/37.34	4(1+3)	215.6	88.2	11.7	115.1	265.5	11.9	11.8	113.8
	608	128.48/37.28	4(0+4)	186.2	-68.9	6.4	753.1	249.2	-4.4	6.9	662.5
Mean	Nsite		5	232.2	-12.4	93.2	1.6	255.5	-2.4	20.8	14.5
Donggo	634	129.02/37.13	8(1+7)	297.1	-65.9	4.6	182.1	270.0	-38.8	4.6	182.5
	925	129.04/37.14	2(2+0)	271.6	-69.8	34.8	53.6	262.6	-27.3	34.7	53.8
	926	129.01/37.14	5(2+3)	317.4	-52.4	4.0	466.8	283.7	-33.1	4.1	456.9
Mean	Nsite		3	299.9	-63.9	21.1	35.1	271.9	-33.4	16.5	57.1

nsp (CPA + Gc) represents the number of specimens (observed directions by principal component analysis and great circles); D'(D), I'(I), α'95(α95) and k1(k2) stand for declination inclination, cone of confidence and precision parameter *in situ* (after tilt correction), respectively.

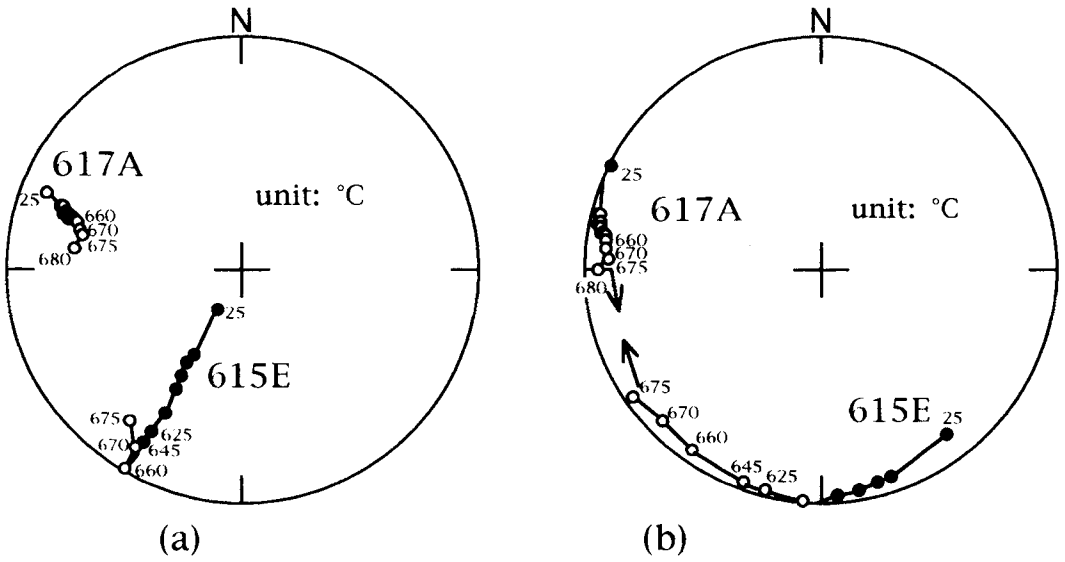


Figure 7. Equal-area projections of the remagnetization great circles for Sites 617 and 615. (a) *in situ* and (b) after tilt correction. Solid and open circles are projections on the lower and upper hemispheres, respectively.

4.1.3 The fold test for Hhc components in the Baegunsan synclinal zone

Sites 617 and 615 are located on the northern and southern limbs of the Baegunsan Syncline (Fig. 1). The timing of the folding is well established as Triassic to Jurassic (e.g. Lee 1987). Two representative great circles (samples 617A and 615E) are shown on the equal-area-projection diagram (Fig. 7). The great-circle trajectories do not converge in the geographic coordinate system (Fig. 7a), while the trajectories converge well in the stratigraphic coordinate system on a direction with a western declination and very shallow inclination (Fig. 7b). This suggests that high-unblocking-temperature (Hhc) components from both sites were acquired before the folding of the syncline.

4.2 High-unblocking-temperature magnetizations from the other Late Carboniferous sites

4.2.1 Site 927

Site 927 is located at the east end of the Baegunsan synclinal zone (Fig. 1a). Seven block samples were collected from red beds of the Manhang Formation, and of these, four samples were successfully cored. Haematite is found to be the only magnetic carrier at this site (Fig. 8b).

As shown in Fig. 8(a) (specimen F), three linear segments were identified by principal-component analysis (Kirschvink 1980). These have high- (Hhc, 660° to 678°C), intermediate- (Hm, 400° to 580°C), and low- (Hl, 200° to 400°C) unblocking-temperature ranges. A considerable drop of intensity

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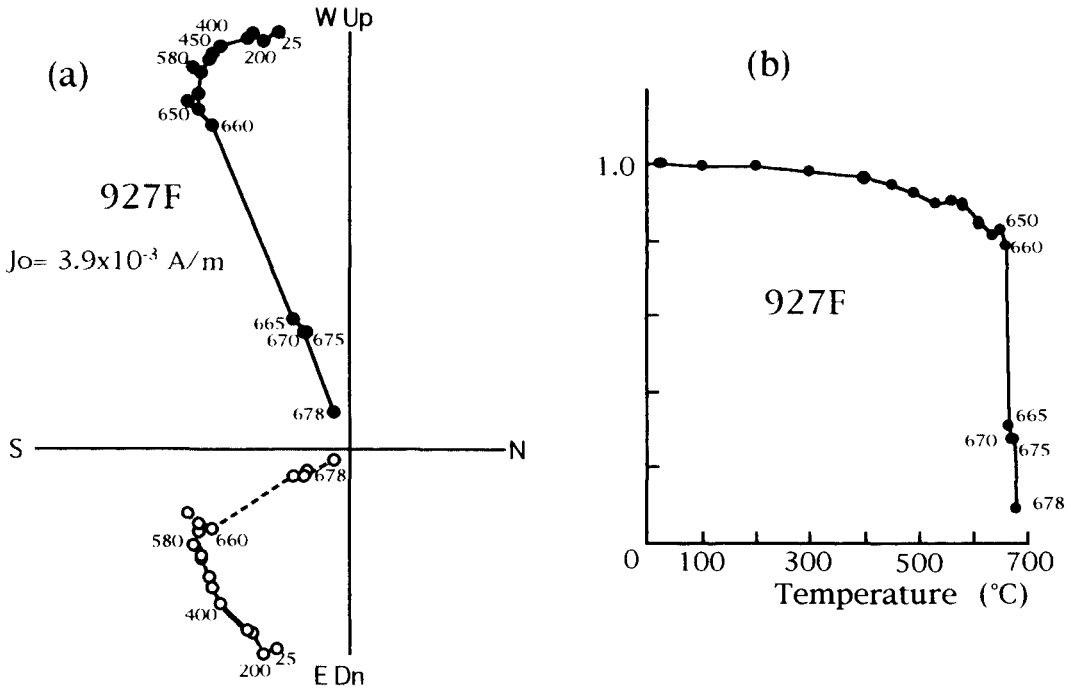


Figure 8. Typical thermal demagnetization results for Site 927. (a) Vector demagnetization diagram and (b) thermal decay of normalized intensity of specimen 927F. See Fig. 4 for explanation of symbols.

between 660° and 678 °C indicates removal of the high-unblocking-temperature component (Fig. 8b). The unblocking-temperature distribution is narrow, as also observed for the Hhc components at Sites 617 (Figs 4b and c) and 615 (Figs 6a and b). The *in situ* site mean direction of the Hhc component was calculated by averaging the directions of the four specimens and is listed in Table 2.

4.2.2 Site 605

Site 605 is located 15 km north of the Yeongweol area (Fig. 1b). Eight block samples were taken from the Moscovian Yobong Formation. Samples D and E were destroyed during laboratory drilling. The demagnetization path varies from specimen to specimen (Fig. 9). Two specimens were rejected, one (C) because of a discrepant direction, the other (G) because of an instability of magnetization above 300 °C.

The remanent magnetization vector (path B in Fig. 9) moves from the initial NRM direction (i) to a cusp point (640 °–650 °C) and then along a great-circle trajectory towards a high-temperature direction (at 675 °C). The movement along the great circle indicates the coexistence of two components of remanent magnetization, and the movement towards the cusp indicates the coexistence of three or more components. A similar demagnetization behaviour is observed for specimen H. After removal of the lower-unblocking-temperature component at 650 °C (trajectory H in Fig. 9), the magnetization vector follows a great-circle trajectory, and approaches the 675 °C direction of specimen B. The demagnetization behaviours imply that specimens B and H contain three components, characterized by high, intermediate, and low unblocking temperatures. The only difference in behaviour between B and H is that the intermediate component in specimen B is much larger than that in specimen H.

In addition, one direct observation of a high-unblocking-temperature component magnetization is obtained from specimen F (670 °–680 °C) and and a great-circle trajectory is obtained from specimen A (600 °–660 °C). The site mean direction for the high-unblocking-temperature component magnetization is obtained using combined analysis (McFadden & McElhinny 1988) of three great circles (specimens A, B, and H) and one direct observation (specimen F). We interpret this characteristic direction as the Hhc component direction at Site 605 (Table 2).

4.2.3 Site 608

Site 608 is located in the lowermost Myobong Formation in the Yeongweol area (Fig. 1b). This site, in which bedding dips at 83°, is thrust against Ordovician rocks that crop out 2 km to the west. Seven block samples were taken from medium-grain-size red sandstones. At this site, a high-unblocking-temperature component could not be obtained directly. Three specimens were discarded, one because of severe remagnetization and the other two because of discordant directions.

Four great-circle trajectories are obtained from magnetization vectors at 645 °–675 °C (specimens E and F), 645 °–679 °C (specimen C), and 645 °–682 °C (specimen B). The characteristic component obtained by the great-circle analysis has a southerly declination and a steep upward inclination in the geographic reference frame. After tilt correction, the site mean direction has a westerly declination and a near-zero inclination (Table 2). We interpret the corresponding characteristic direction as the Hhc component direction at Site 608.

4.3 High-unblocking-temperature magnetizations from the Permo-Triassic sites

Five red-bed sites of the Permo-Triassic Donggo Formation were sampled at the eastern end of the Baegunsan synclinal

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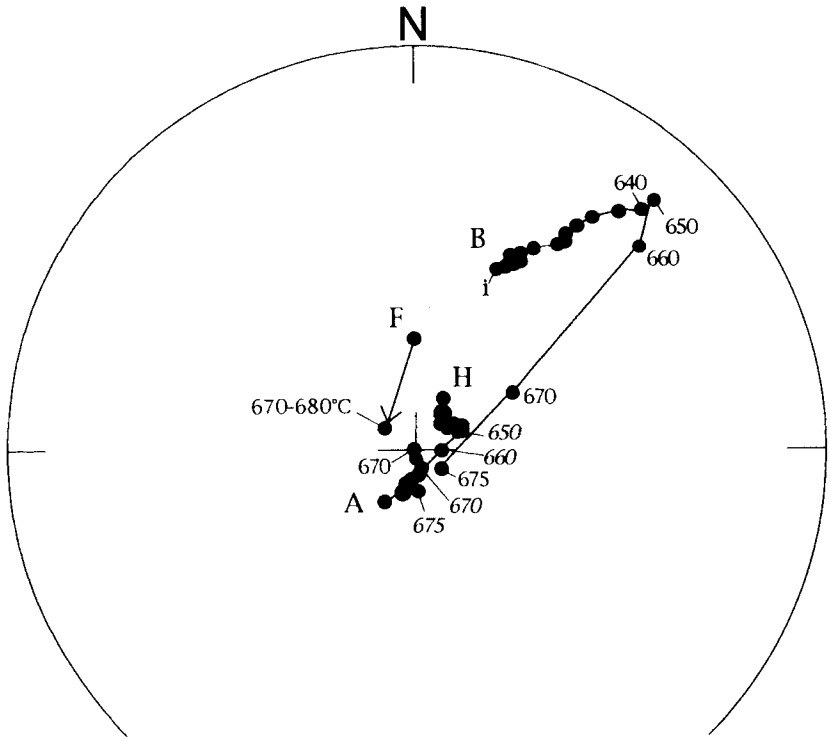


Figure 9. Equal-area projection of thermal demagnetization results for Site 605. The numbers in italics denote demagnetization temperatures for specimen H. See Fig. 5 for explanation of symbols.

area in the Okcheon zone (Fig. 1a and Table 1). All sites are located on the same limb of the syncline.

4.3.1 Site 634

Red sandstones were sampled across a 3 m stratigraphic thickness (Fig. 1a). As shown in Fig. 10(a), a direct observation in the temperature range 679° to 685 °C for the stratigraphically lowest specimen A isolates a high-unblocking-temperature component with a downward inclination. Great-circle trajectories, after removal of the soft component from 400° to 600 °C, are shown for seven specimens in Fig. 10(b). The remanent magnetization of specimen A has a positive inclination (Fig. 10a), while those of the other specimens have negative inclinations. We interpret the remanent magnetization of specimen A as recording a reversed polarity relative to the other seven specimens. For comparison of directions we therefore inverted the polarity of the remanent magnetization of specimen A (square A in Fig. 10b). The stable endpoints for seven great-circle trajectories are shown as triangles (Fig. 10b). The high-unblocking-temperature component magnetization is obtained by the McFadden & McElhinny (1988) method (star M in Fig. 10b). After tilt correction, the high-unblocking-temperature component has a moderate inclination (Table 2). This direction is distinct from that of Hhc components of Late Carboniferous red beds, which have low inclinations. We hereafter call this high-unblocking-temperature component magnetization of Permo-Triassic red beds the Hhpt component.

4.3.2 Site 925

Seven block samples were collected from reddish shales (Fig. 1a and Table 1). Specimens were obtained from only two samples (A and C) because of very high fissilities. Two specimens directly reveal magnetizations with unblocking temperatures above 675 °C. The intensity decreases drastically in the last five steps between 669° and 679 °C. The direction for the high-unblocking-temperature component magnetization (*in situ*) has a northwest declination and a high upward inclination (Table 2).

4.3.3 Site 926

Eight block samples were collected from red sandstones (Fig. 1a). Two of those (D and E) were lost during coring because of high fissility. One specimen (F) was rejected, because of unstable magnetization above 660 °C. The high-unblocking-temperature component was obtained directly from specimens B and C between 665° and 673 °C. Two direct observations were combined with three great-circle trajectories (specimens A, G, and H) for calculation of the site mean direction (Table 2).

4.4 The other sites

We could not find any high-unblocking-temperature components at the following six sites: Site 633 of the Manhang Formation; Sites 604, 606, and 607 of the Myobong Formation; and Sites 923 and 924 of the Donggo Formation (Fig. 1). The NRM directions were discarded for one or more of the following reasons:

(1) suspicion of remagnetizations;
(2) no systematic demagnetization behaviour
(3) less than three great circles or less than two directly obtained directions from a site

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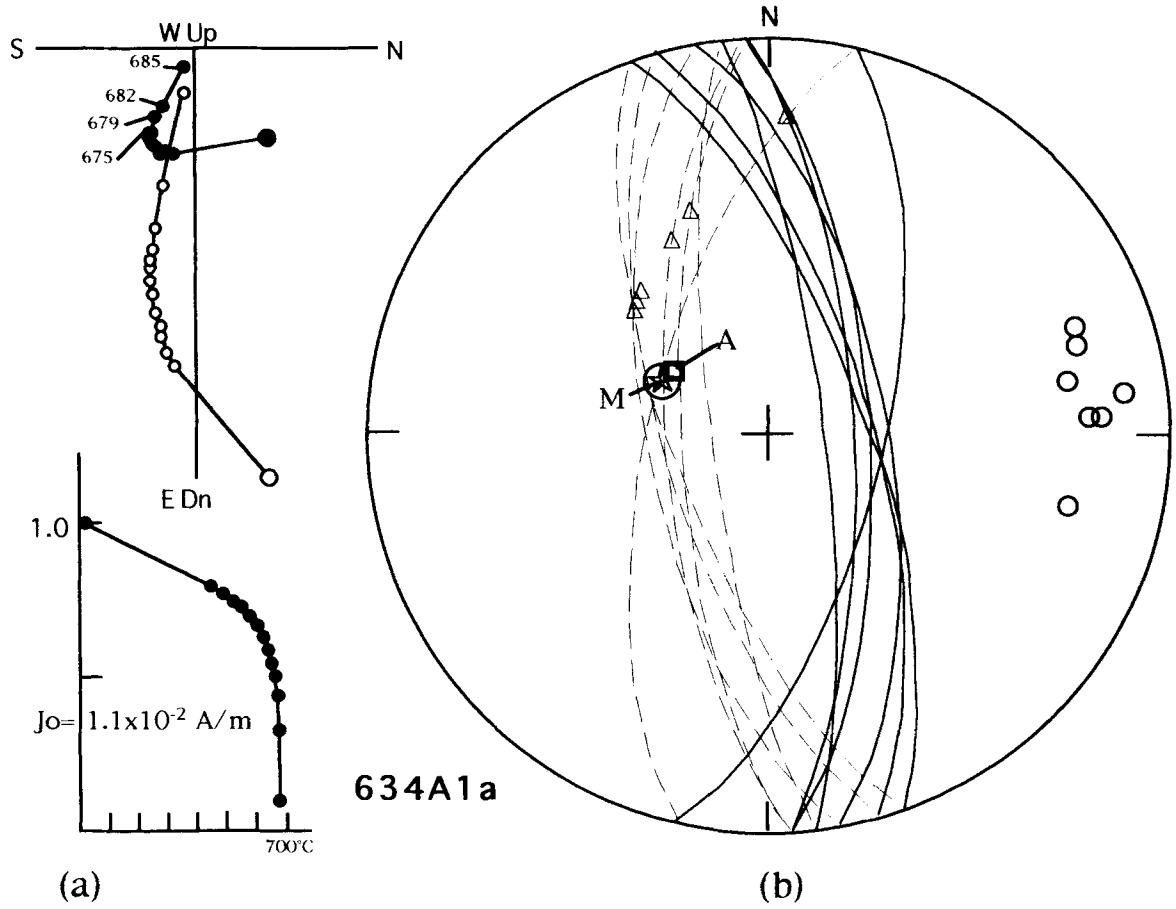


Figure 10. Thermal demagnetization result for Site 634. (a) Vector demagnetization diagram for the direct observation. For symbol explanation, see Fig. 4 (a). (b) Combined analysis of a direct observation (634A) and seven great circles. Star (M) and oval represent the Site 634 mean direction and its cone of confidence obtained from the direct observation at 634A, but its polarity is inverted. Triangles and circles indicate stable endpoints and poles to great-circle trajectories, respectively. Open symbols represent projections on the upper hemisphere.

(4) weak NRM intensity
(5) hard VRM acquired during demagnetization.

Regional-scale thrusts and faults are located 2 to 3 km west of the Yeongweol sampling area (Sites 604, 606, and 607). The tectonism probably remagnetized the rocks. In addition, a Cretaceous rhyolite (labelled 'volcanics' in the legend in Fig. 1) crops out near Sites 923 and 924. The stable components at Site 923 are identical to the Cretaceous direction reported for the Korean Peninsula (Lee *et al.* 1987). The initial NRM intensities of the specimens of pale-grey sandstone from Site 633 range from 2.1 to 3.0×10^{-3} A m^{-1}, and thermal demagnetization show directions that are identical to the geocentric axial dipole field direction.

5 PETROGRAPHIC OBSERVATIONS: THE RELATIONSHIP BETWEEN MAGNETIZATIONS AND HAEMATITE GRAIN TYPES

Petrographic observation was carried out on 25 thin sections and eight polished sections from red-bed samples of the Pyeongan Supergroup. Three types of haematite were seen under reflected and transmitted light (Fig. 11). The grains in the red beds comprise quartz, feldspar, lithic fragments, tourmaline, and zircon. The matrix is generally altered to sericite, haematite, and clay minerals.

Type A is a detrital specular haematite, generally tens of micrometres in grain size. The detrital grains are subrounded to rounded, and sometimes show shear dislocations. Type A grains are occasionally surrounded by elongated, non-detrital specular haematites (type B), or are sometimes overgrown by pigmental haematite (type C). These features indicate that type A haematites were formed earlier than types B and C. Many of the type A grains contain thin (<2 μm), curved lamellae of ilmenite, and pseudomorphs (martite) after magnetite, which suggests that this kind of haematite was formed by oxidation of magnetite (Fig. 11a). This indicates that the martites may make an important contribution to the remanent magnetization of the primary component. The timing (pre-depositional or post-depositional) of the oxidization is not clear from petrographic observation. In the case of a pre-depositional origin, the haematite grains would have been oxidized during or before transportation from the source area, and the remanent magnetization would therefore be ascribed to DRM or post-DRM, while for a post-depositional origin, the oxidation may have taken place during diagenesis, and the remanent magnetization would be ascribed to CRM. We

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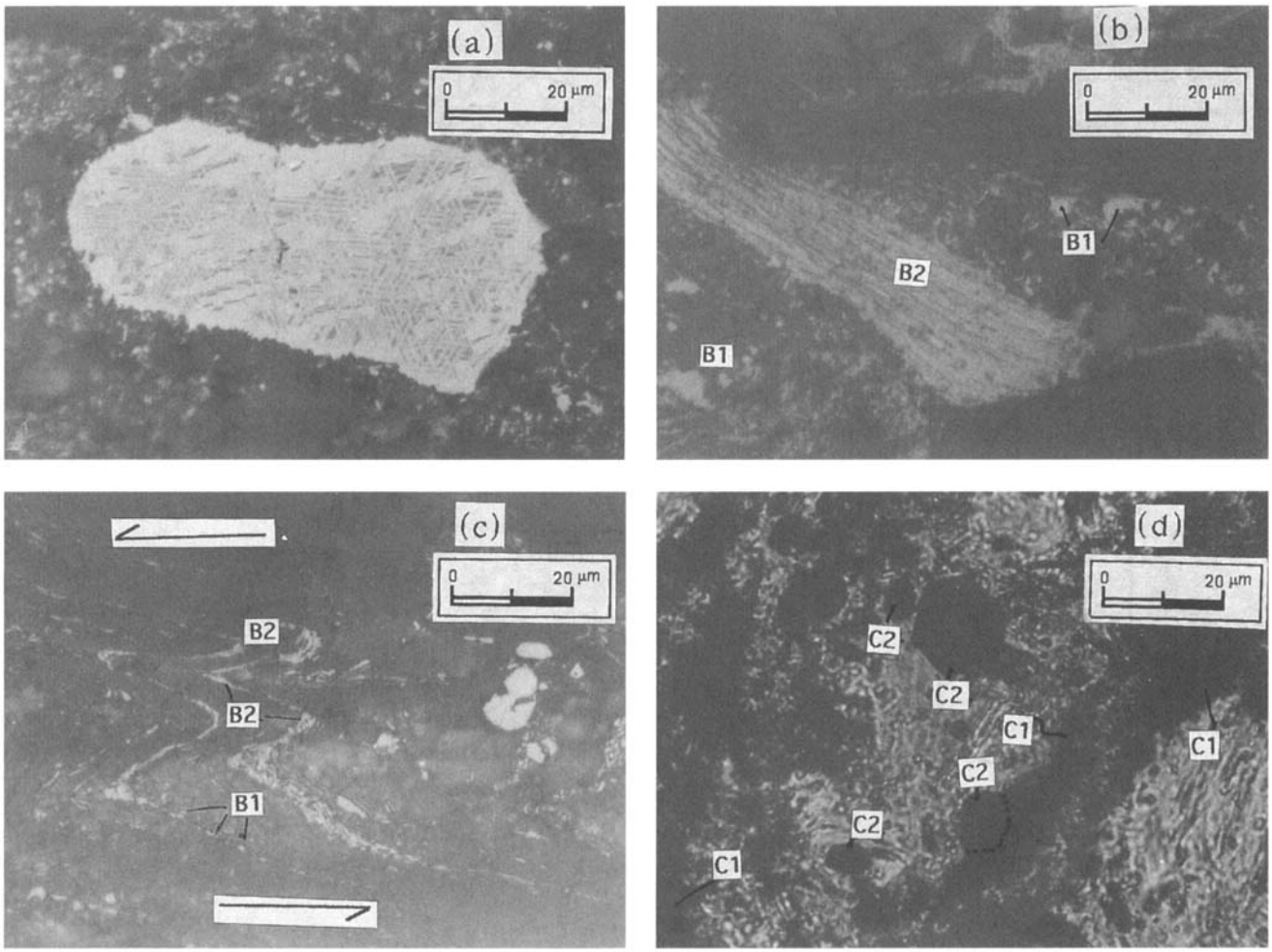


Figure 11. (a) Thin-section photo-micrograph under reflected light (sample 617H). The detrital specularite (type A) showing martitization of a titanomagnetite grain through haematite (bright)–ilmenite (grey) lamellae. (b) Thin-section photomicrograph under reflected light (sample 617H). Type B2 showing a mica-shaped grain replaced by specular haematite. The fine-grained bright particles represent microcrystalline non-detrital specularite (type B1) in the matrix. (c) Thin-section photomicrograph under reflected light (sample 617K, reddish shale). The aggregations of types B1 and B2 grains show alignment of micro-drag-fold traces of an incompetent layer during folding. Arrows indicate the direction of deformation between competent sandstone layers. (d) Thin-section photomicrograph under transmitted light (sample 617L). Type C showing pigmental haematites (black crystals or spots on photograph, but red-translucent in reality). Some crystalline pigmental haematite of type C2 (e.g. hexagonal cross-sections, lower centre) are larger than 10 μm. Type C1 pigmental haematite, however, is of a very *fine* amorphous nature.

believe that the remanent magnetization carried by type A grains might be correlated with the characteristic Hhc (Hhpt) component.

Type B is defined as a non-detrital specularite, and is obviously of secondary origin. This type of haematite forms fine aggregates (type B1, <2 μm), or tabular sheets (type B2, 10–100 μm). Type B1 fills small cavities, replaces fine-grained clay minerals, or coats detrital grains. Because of its fine-grained nature, it is not clear if type B1 carries a remanent magnetization. Type B2 replaces pre-existing silicate minerals, such as sericite, forming pseudomorphs that contain cleavage traces (Fig. 11b). Non-detrital specularite grains (types B1 and B2) line micro-drag-fold traces of incompetent layers (Fig. 11c), indicating that grains of both types B1 and B2 were formed during a folding event. Walker *et al.* (1981) suggested that type B grains formed by replacement are the product of precipitation of iron released by intrastratal alteration of pre-existing, unstable, iron-bearing detrital materials. Grains of type B2 are large enough to carry a remanent magnetization, and could be responsible for acquisition of the observed secondary CRM (Hm component) during the Early Mesozoic Songrim and Daebo Orogenies (Lee 1987).

Type C haematite occurs as pigmental aggregations. As shown in Fig. 11(d), this haematite is generally fine-grained and amorphous (type C1, <2 μm), or sometimes euhedral crystalline (type C2, 10 to 30 μm). Type C grains are translucent red in transmitted light, and show poor reflectivity. They form overgrowths, or sometimes penetrate specularite grains (types A and B). Type C is the principal pigmenting material in the red beds, and is probably related to dissolution of specular haematite grains or the matrix (Walker *et al.* 1981). Butler (1992) suggested that they may have formed by post-depositional chemical processes, such as dehydration of ferric oxyhydroxides or alteration of Fe-bearing silicates. Euhedral pigmental haematite (type C2) is less dominant than the non-detrital specularite of tabular sheet form (type B2). The large euhedral grains of type C2, which we interpret as being authigenic, have single to pseudo-single domain sizes, and may

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be responsible for the secondary CRM. However, most of the amorphous pigmental haematite of type C1 may not contribute to the remanent magnetization because of its fine-grained nature. In the present study, the Hl component has shown a normal polarity present field direction, and unblocking temperatures of 450° to 660 °C. In the Okcheon zone, no post-Tertiary igneous activity has been reported. Such high unblocking temperatures cannot be explained by the relationship between relaxation time and blocking temperature, as suggested by Pullaiah *et al.* (1975). Jackson (1990) proposed that the size dependence of thermoviscous magnetization may not follow Pullaiah *et al.*'s equation, which is based on Neel's single-domain theory. Therefore, the two possible mechanisms of the Hl remanent magnetization component are a CRM due to euhedral pigmental haematite (type C2) grains and a VpTRM (viscous partial thermoremanent magnetization) in specularites (types A and B2).

We should note that the observed grain sizes for the three types of haematite are not necessarily the magnetic domain sizes. For example, most type A haematite grains contain many narrow ilmenite lamellae. In this case, the domain size is the grain size divided by the number of lamellae. Most of these grains are possibly of single to pseudo-single domain size since single-domain haematite grain can be larger than 15 μm (Banerjee 1971) for an elongate grain. This implies that two types of specularite (types A and B2) carry the Hhc (Hhpt) and Hm components of remanent magnetization. This is supported by the Hhc (Hhpt) and Hm component magnetizations being thermally stable up to the Curie temperature of haematite. The spectra, especially of the primary (Hhc or Hhpt) component magnetizations, break down in a narrow temperature range of 660 °–685 °C (e.g. Figs 4b, c, 8b, etc.).

6 RESULTS AND DISCUSSION

6.1 Thermal demagnetization of the multicomponent remanences

Roy & Park (1972) performed thermal demagnetization and chemical leaching experiments on red beds of the Hopewell Group, which contain one detrital and two types of chemically precipitated haematite. They claimed that one CRM was removed by the thermal demagnetization (CRM_B: red material), but DRM and the other CRM (CRM_A: specularite) could not be separated thermally. A combined thermal demagnetization and chemical leaching technique was effective in separating DRM and CRM_A (see also Roy & Lapointe 1978).

We find that the three kinds of haematite-component magnetizations can be separated in red beds of the Pyeongan Supergroup by careful thermal demagnetization analysis. We believe that the primary Hhc (Hhpt) component magnetization from the red beds is carried by detrital specularite (type A), composed of more than 80 per cent martite. We are not sure if the DRM and CRM_A from the Hopewell Group could be separated by detailed thermal demagnetization; however, we suggest that the Hopewell Group and other remagnetized red beds should be re-examined through careful and detailed thermal demagnetization, because a combined thermal demagnetization and chemical leaching technique is time consuming.

6.2 Late Carboniferous palaeomagnetic directions in the Okcheon zone

Late Carboniferous palaeomagnetic directions (Hhc component) were obtained from two adjacent Baegunsan (the Manhang Formation) and Yeongweol (the Yobong Formation) areas (Fig. 1). We obtained five site mean directions for the characteristic Hhc component from nine sampling sites (Table 2). Westward (for 22 specimens) and eastward (for one specimen) declinations with near-zero inclinations indicate reversals. We conclude that the polarity with westward declination is probably reverse because of a predominant polarity reversal in Permo-Carboniferous times, as expressed by the Permo-Carboniferous reverse superchron.

According to previous geological studies, the Baegunsan and Yeongweol areas differ greatly in stratigraphy and structural character (Lee 1987). However, the remanent directions of the Manhang Formation (Baegunsan area) are not distinguishable from those of the Yobong Formation (Yeongweol area). This implies that these basins were originally adjacent, and that no palaeomagnetically detactable relative motion has occurred between them since Moscovian time.

The formation mean direction for the Late Carboniferous was calculated with Fisher's (1953) statistics. The mean direction for the Hhc component from the Late Carboniferous red beds is D = 255.5°, I = −2.4°, and $\alpha_{95} = 20.8°$ in stratigraphic coordinates (Table 2, Fig. 12b). The ratios of precisions and α_{95} angles of the mean directions of the Hhc component after/before tilt correction are $k_2/k_1 = 9.06$ and $\alpha_{95}/\alpha'_{95} = 0.22$, indicating that the Hhc component was acquired at, or close to, the time of deposition of the red beds. From this direction, the palaeomagnetic pole for Late Carboniferous is calculated as 226.8°E, 12.2°N ($\alpha_{95} = 15.5°$). The Late Carboniferous palaeo-

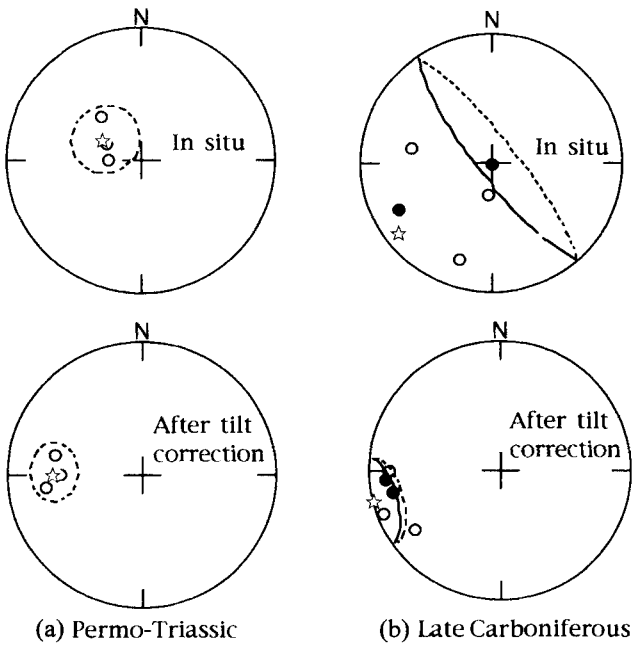


Figure 12. Site mean directions (stars) of *in situ* (upper diagrams) and after tilt correction (lower diagrams) for the Hh component magnetizations for (a) Permo-Triassic and (b) Late Carboniferous rocks. Open and solid circles are projections on the upper and lower hemispheres, respectively.

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latitude for the site calculated from the inclination is 1.2°N, i.e. the Okcheon zone was located near the equator.

6.3 Permo-Triassic palaeomagnetic direction in the Okcheon zone

Three characteristic site mean directions of the Hhpt component were obtained from five sampling sites in the Baegunsan area. The formation mean direction of the Hhpt component in Table 2 is interpreted as the primary direction for the following reasons.

1. The increase in the precision parameter after tilt correction ($k_2/k_1 = 1.63$ and $\alpha_2/\alpha_1 = 0.78$).
2. The inclination of the Hhpt component (−33.4°) is different from the Hhc component of Late Carboniferous origin (−2.4°).
3. Neither *in situ* nor tilt-corrected directions are identical to any post-Palaeozoic directions. Remagnetization is thus unlikely.

The mean direction of the Hhpt component is D = 271.9°, I = −33.4° ($\alpha_{95} = 16.5°$, Table 2 and Fig. 12a). The statistical confidence levels (α_{95} and k) improve after tilt correction, and the direction is easily distinguishable from the Late Carboniferous one (D = 255.5°, I = −2.4°). This suggests acquisition of the Hhpt-component magnetization at or close to the time of deposition of the red beds. The corresponding Permo-Triassic palaeomagnetic pole is calculated as 203.1°E, 9.4°N ($\alpha_{95} = 19.7$).

6.4 Comparisons with previous results from the Okcheon zone

As clearly shown by the present study, the red beds of the Pyeongan Supergroup contain multiple haematite-magnetization components. Rock-magnetic studies are indispensable for obtaining correct characteristic directions. In most cases, the unblocking temperature of the secondary Hl component ranges from 550° to 650 °C. In addition, the maximum unblocking temperature of the secondary Hm component unblocking temperature spectrum overlaps that of the primary Hhc component above 680 °C. Insufficient detail in demagnetization above 670 °C may lead to serious errors in the determination of a characteristic direction, and the Hhc component cannot be obtained directly.

Three previous results from Late Carboniferous and Permo-Triassic red beds are shown in Fig. 13 and listed in Table 3. Two Permo-Triassic results were reported from the red beds of the Donggo Formation in the Okcheon zone by Shibuya *et al.* (1988) and Kim & Van der Voo (1990). Kim & Van der Voo (1990) determined a high-temperature component magnetization, and obtained its site mean direction using the combined great-circle and stable-endpoint method (Bailey & Halls 1984). However, critical demagnetization data above 660 °C were not determined. Since each great circle was defined from a limited set of observations (less than 670° or 660 °C), their site mean seems less reliable than ours. McFadden & McElhinny (1988) argued that the Bailey & Hall (1984) method can cause bias, particularly when direct observations are not available. Further bias may occur when the great circles are defined on the basis of too few high-temperature demagnetization steps.

Two Late Carboniferous palaeomagnetic directions from contemporaneous red bed formations in the Okcheon zone were obtained by Shibuya *et al.* (1988) and Kim *et al.* (1992). These are shown in Table 3 and Fig. 13(b). Two site mean directions, from Jangseong (Site 43 in Shibuya *et al.* 1988) in the Baegunsan area and Mungyeong (Site 20 in Shibuya *et al.* 1988) in the western region of the Okcheon zone, were provided by Shibuya *et al.* (1988). The site mean direction from the Mungyeong area should be re-examined, because our new results (Site 644, three direct observations with dual polarities) from the same outcrop as Site 20 (in Shibuya *et al.* 1988) give a different direction (D = 155.8°, I = −3.7°, $\alpha_{95} = 12.8$, and $k =$ 93.6 in the stratigraphic reference frame). For this reason only the site mean direction from the Jangseong area (Site 43 in Shibuya *et al.* 1988) is discussed here.

As shown in Figs 13(a) and (b), the confidence circles do not overlap for the mean directions of the present and the two previous studies. These discordances could have occurred by contamination with secondary (Hm and/or Hl) components or viscous magnetization acquired during measurements.

Recently, Doh & Piper (1994) reported high-blocking-temperature (hbt) components from Late Palaeozoic sedimentary rocks in the Baegunsan syncline of the Okcheon zone. Their hbt directions are widely scattered between sites (Figs 4b, 8b, and 9 in Doh & Piper 1994). They suspected that components of unresolved local rotation contribute to the dispersion, and omitted three out of seven site mean directions for the Triassic (Permo-Triassic in the present study) Donggo Formation and seven out of 10 site mean directions for the Carboniferous Manhang Formation on the basis of less anti-parallel population (e.g. Table 4 and Fig. 9 in Doh & Piper 1994). They interpreted the blocks to have undergone at least two rotation events, during the Late Jurassic–Early Cretaceous and the Cretaceous–Recent. From their interpretation, it should be noted that non-deformed versus deformed sites (during the Late Jurassic–Early Cretaceous) obtained from the hbt component (Tables 2, 3 and 4, and Figs 4 and 9 of Doh & Piper 1994) are randomly distributed throughout the synclinal zone (for example, Sites 18 and 39 versus Site 40, Site 23 versus Sites 21 and 22, or Site 32 versus Sites 33 and 35 in Fig. 1 of Doh & Piper 1994). Geologically, this analysis is not acceptable because the presence of a simple synclinal structure is well established for the Baegunsan synclinal zone, and the Palaeozoic formations or their key beds are well traced, with a similar bedding trend throughout the zone (Lee 1987). Most of the hbt components could be pervasively contaminated by secondary (Hm and/or Hl) components or with viscous magnetization acquired during measurements, because they demagnetized the samples every 20 °C above 600 °C. Moreover, Doh & Piper suggested a Cretaceous–Recent clockwise block rotation of the Baegunsan syncline in the Okcheon zone, relative to the Korean Peninsula, by about 30° (Lee *et al.* 1987), on the basis of their site mean overprint direction in the Baegunsan synclinal zone. However, there are no previous palaeomagnetic data or geological evidence indicating a Cretaceous–Recent local rotation event in the Okcheon zone (Lee 1987).

The previously estimated Permo-Triassic and Late Carboniferous poles should be discarded for the following three reasons. First, they were not based on detailed thermal

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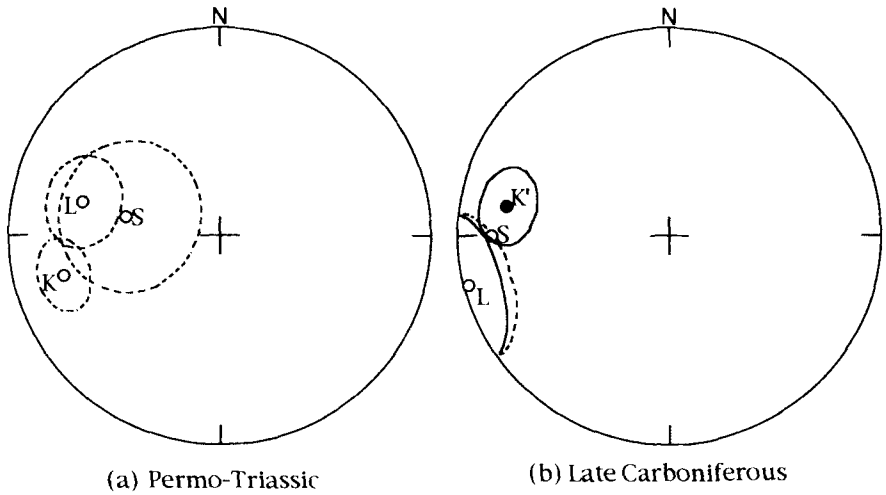


Figure 13. Site mean directions (after tilt correction) for (a) Permo-Triassic (listed in Table 3) and (b) Late Carboniferous (listed in Table 3) rocks from the Okcheon zone, Korea. L: present study; S; Shibuya *et al.* (1988); K: Kim & Van der Voo (1990); K': Kim *et al.* (1992). Open and solid circles are projections on the upper and lower hemispheres, respectively.

Table 3. Comparisons of previous and present palaeomagnetic results from Late Carboniferous (lower) and Permo-Triassic (upper) rocks in the Okcheon zone. Abbreviations as given in Table 2.

References	Nsite	D'	I'	α'_{95}	k1	D	I	α_{95}	k2
this study	3	299.9	-63.9	21.1	35.1	271.9	-33.4	16.5	57.1
Shibuya et al.(1988)	3	304.5	-41.9	26.2	23.2	282.0	-52.6	28.8	19.4
Kim & Van der Voo(1990)	4	251.5	-24.1	-	-	255.1	-2.4	12.1	27.0
this study	5	232.2	-12.4	93.2	1.6	255.5	-2.4	20.8	14.5
Shibuya et al.(1988)	1	261.5	27.3	-	-	269.4	-17.0	6.9	-
Kim et al.(1992)	5	270.6	57.8	44.7	5.1	280.5	22.5	13.0	17.0

demagnetization between 660° and 680 °C, as discussed above. Second, they were obtained without precautions against the acquisition of viscous magnetization. We found that it took several tens of minutes for viscous components induced after heating above 660 °C to decay below the noise level in the cryogenic magnetometer. Third, the possibility of selection of the demagnetization range for determining great circles was inadequate. In the red beds of the study area, the Hhc (Hhpt) component is seriously contaminated by the intermediate- (Hm) and low- (Hl) unblocking-temperature haematite component magnetization (see Fig. 4). In this case, directions of Hl component magnetization should be completely excluded from the remagnetization great circles.

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